

Article

Catalytic Coupling of Oxidative Phosphorylation, ATP Demand, and Reactive Oxygen Species Generation

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ABSTRACT Competing models of mitochondrial energy metabolism in the heart are highly disputed. In addition, the mechanisms of reactive oxygen species (ROS) production and scavenging are not well understood. To deepen our understanding of these processes, a computer model was developed to integrate the biophysical processes of oxidative phosphorylation and ROS generation. The model was calibrated with experimental data obtained from isolated rat heart mitochondria subjected to physiological conditions and workloads. Model simulations show that changes in the quinone pool redox state are responsible for the apparent inorganic phosphate activation of complex III. Model simulations predict that complex III is responsible for more ROS production during physiological working conditions relative to complex I. However, this relationship is reversed under pathological conditions. Finally, model analysis reveals how a highly reduced quinone pool caused by elevated levels of succinate is likely responsible for the burst of ROS seen during reperfusion after ischemia.

INTRODUCTION

The *in vivo* rate of oxidative ATP synthesis is matched to the cellular ATP hydrolysis demand in working muscles through a feedback mechanism involving ATP hydrolysis products, inorganic phosphate (P_i), and ADP, as a result of increases in cardiac work rate (1–6). The *in vivo* concentrations of these metabolites observed at different work rates have been shown to closely match the concentrations required to produce the corresponding rates of ATP synthesis in purified mitochondria (7–10). In the working heart, the concentration of cytoplasmic P_i is thought to be the most important feedback signal controlling oxidative phosphorylation (1–3,11).

The fundamental feedback relationship between products of ATP hydrolysis and stimulation of ATP synthesis *in vivo* has been challenged by an alternative hypothesis whereby the open-loop stimulation by calcium, or some other signal, represents a significant contribution to *in vivo* control of oxidative ATP synthesis (12–14). While it is possible to design *in vitro* experiments that demonstrate the effects of calcium on mitochondrial NADH production, respiratory activity, and ATP synthesis (15,16), attempts to demonstrate these stimulatory effects within the physiological range of calcium concentrations, temperature, ionic strength, and substrate concentrations, have failed (7,17,18). Furthermore, it has been demonstrated that calcium transport is not necessary for energy homeostasis in heart (19).

In addition to reducing oxygen to water via complex IV, the other mitochondrial respiratory complexes are capable of producing a number of reactive oxygen species (ROS)

(20). It was originally believed that the ROS-generating side reactions were purely detrimental. For instance, many proteins possess cysteine and methionine residues that are sensitive to ROS levels (21–23), and certain phosphatases may be particularly susceptible to oxidative stress induced by elevated ROS (24). However, it is now understood that certain reactive oxygen species are important physiological signaling molecules (25,26). Elevated ROS levels from complex I caused by matrix alkalization after mitochondrial KATP channel opening contributes to cardioprotection against ischemia/reperfusion injury (27). However, we should note that the existence of the KATP channel is still debated (28). Activation of the PI3K-Akt pathway leads to increased ROS production (29). Reactive oxygen species also play a role in regulating transcription of genes involved in energy metabolism (30) by stimulating PGC1 α (31).

While it is accepted that mitochondria are a significant source of cellular ROS, the precise sites of ROS generation (specifically hydrogen peroxide and superoxide ion) in mitochondria have only recently been determined (32–34). The sites of ROS production are substrate-specific and some sites have a higher capacity for ROS production than others (33). For example, when mitochondria oxidize succinate *in vitro*, the major source of ROS is believed to be the quinone-reductase site in complex I, but the NADH-oxidase site also likely contributes to ROS production (35,36). When glutamate and malate are oxidized, the NADH-oxidase site in complex I and the quinol-oxidase site in complex III are the major sites of ROS production. In addition, dihydrolipoamide dehydrogenase, also known as E3, has the capacity to produce a significant amount of ROS during substrate oxidation (37–40). Extrapolating ROS production rates measured *in vitro* to expected rates

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in vivo must be done with caution (26). As it is currently extremely difficult to accurately measure ROS production in vivo, a mathematical model is required to identify the precise conditions that lead to ROS production from the various potential production sites and predict how these sites will contribute to net production under physiological and pathological conditions.

Our recent work identifying and analyzing thermodynamically constrained models describing the catalytic mechanisms of respiratory complexes I (35) and III (41) as well as the availability of the data of Vinnakota et al. (42) on cardiac mitochondrial respiratory rates, redox states of NADH and cytochrome *c*, and membrane potential under physiological concentrations of ADP and P_i , provide the opportunity to develop a theoretical model of oxidative phosphorylation of unprecedented biophysical detail and robustness. Because the kinetics of superoxide and hydrogen peroxide production by complexes I and III have been determined for the models of Bazil et al. (35,41), the present objective is to integrate these mechanisms into a model that can simulate both the respiratory dynamics associated with ATP production and the kinetics of ROS production in a single integrated system.

The developed model reveals how oxidative ATP synthesis responds to changes in ATP demand through a negative-feedback control mechanism involving both ADP and P_i , predicts that an increase in the quinone pool (Q pool) redox state is responsible for the apparent activation of complex III by inorganic phosphate, and shows that complex III is responsible for more ROS production during physiological working conditions compared to complex I but that this relationship is reversed during pathological conditions. Finally, the model is used to analyze a recently reported phenomenon in which highly reduced Q pool caused by elevated levels of succinate can be responsible for the burst of ROS seen during reperfusion after an ischemic or hypoxic period (43).

MATERIALS AND METHODS

The 2005 model of oxidative phosphorylation of Beard (44), which uses a mixture of kinetic and mass action descriptors for the essential components, provides a basic framework for model development. A model diagram of the updated model is shown in Fig. 1. The model descriptors for NADH production, complexes I, III, and IV were updated to capture the substrate/product and hydrogen peroxide/superoxide production kinetics (35,41,45). An empirical descriptor for complex II that is proportional to the NADH production rate was also added to the model. In the original Beard model (44), the NADH production descriptor contained a P_i activation mechanism. This was replaced with a regulatory function that incorporates feedback in mitochondrial [ATP], [ADP], and [P_i]. See Vinnakota et al. (42) for discussion on the development of this model component. The complex I model includes the components necessary to simulate NADH-quinone oxidoreductase activity as a function of pH and mitochondrial membrane potential ($\Delta\Psi$) and also the detailed redox biochemistry required to simulate hydrogen peroxide and superoxide generation by the FMN and the SQ sites (35). In a similar manner, the complex III model simulates quinol-cytochrome *c* oxidoreductase activity as a function of pH and $\Delta\Psi$, as well as the redox biochemistry involved in superoxide formation

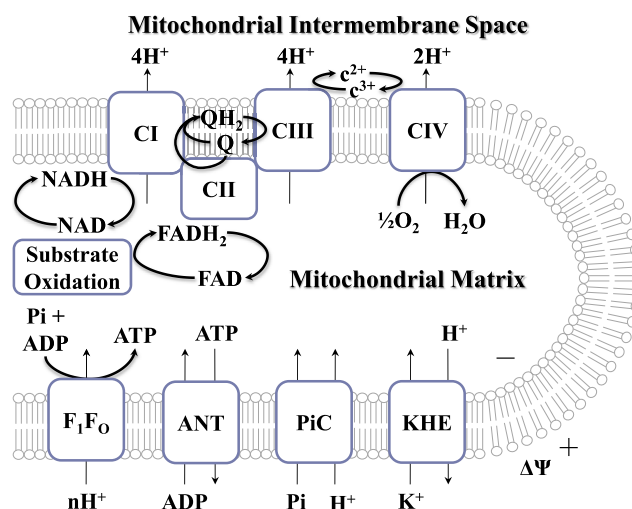


FIGURE 1 Model diagram. The model consists of descriptors for substrate oxidation, the electron transport chain, oxidative phosphorylation, and the potassium/proton exchanger (KHE). The substrate oxidation descriptor is an empirical function characterizing NADH and $FADH_2$ production. The electron transport chain consists of NADH-ubiquinone oxidoreductase (CI), succinate dehydrogenase (CII), ubiquinol cytochrome *c* oxidoreductase (CIII), and cytochrome *c* oxidase (CIV). The oxidative phosphorylation components are ATP synthase (F_1F_0), adenine nucleotide translocase (ANT), and the inorganic phosphate carrier (P_iC). The number of protons required to produce one ATP molecule (n) by F_1F_0 is set to 2.67. An extra proton enters mitochondria through the P_iC and ANT, which raises the H:ATP ratio to 3.67 as seen from outside the mitochondria. To see this figure in color, go online.

(41). Hydrogen peroxide and superoxide scavenging is modeled using a simple empirical description using published estimates of the first-order rate constants for the superoxide dismutase and peroxidase reactions (46). The complex IV model captures the relationship between O_2 consumption and the major kinetic drivers, the concentration of cytochrome *c*²⁺ and $\Delta\Psi$, as inferred by Murphy and Brand (45). The model for adenine nucleotide translocase (ANT) was updated with more complete model of ANT kinetics (47). Lastly, an ATPase model was added to enable simulation of the Vinnakota data set. The full model equations and parameter details are given in the Supporting Material and are provided in the MATLAB codes (The MathWorks, Natick, MA) for simulating the model.

The data used to parameterize the model provide steady-state measurements of complex IV flux, NADH and cytochrome *c* redox states, and inner mitochondrial membrane potential as functions of extramitochondrial [ADP] and [P_i] maintained at different levels by the addition of the ATP hydrolyzing enzyme used to drive suspensions of purified mitochondria into different work states. Full experimental details are reported by Vinnakota et al. (42). In brief, isolated rat heart mitochondria were subjected to different ATP demand loads in the presence of low (1 mM) or high (5 mM) P_i . These data are used in conjunction with supplemental data to estimate the model parameters (see the Supporting Material).

The flux control coefficients were numerically computed by evaluating the following expression:

$$C_k^j = \frac{dJ_i}{dp_k} \frac{p_k}{J_i} \quad (1)$$

where C_k^j is the i th flux control coefficient for the k th reaction, J_i is the i th flux, and p_k is the k th parameter associated with the k th reaction (usually an enzyme total or activity). The flux control coefficients are normalized

sensitivity coefficients and do not necessarily reflect physiological control mechanisms. They are a measure used to assess the relative importance of a system component for a given system output.

The model was developed, parameterized, and simulated on an HP desktop PC with an Intel i7-3770 CPU at 3.40 GHz and 16.0 GB of RAM using MATLAB ver. 2014b (The MathWorks). A local, gradient-based optimization algorithm was used to estimate the model parameters.

RESULTS AND DISCUSSION

Our model of mitochondrial respiration and oxidative phosphorylation is depicted in Fig. 1. This model was based on the 2005 Beard model (44) and consists of new descriptors for substrate oxidation, the electron transport chain, and oxidative phosphorylation. The model was parameterized using a data set that elucidates how oxidative ATP synthesis is regulated in the heart under physiological conditions (42). Model simulations compared to data are shown in Fig. 2. To simulate the experiments of Vinnakota et al. (42), the extramitochondrial ATPase rate is adjusted to simulate the steady-state relationships between observed variables

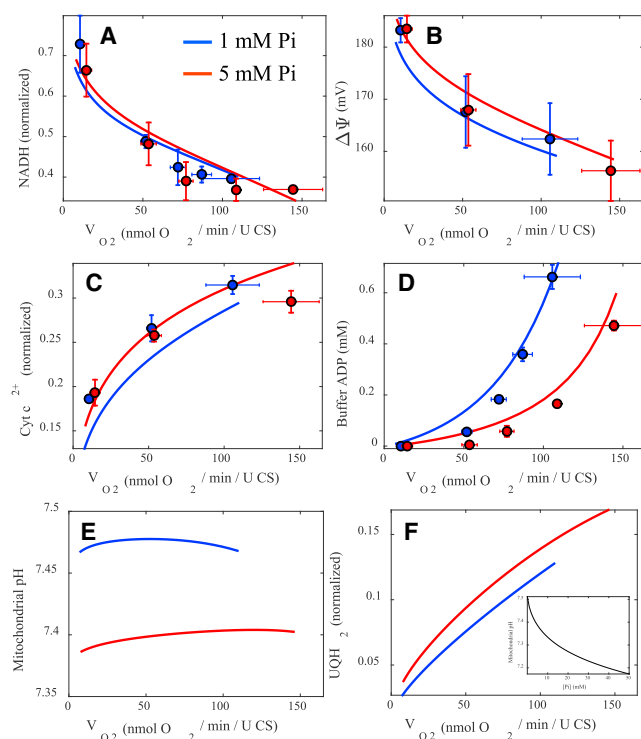


FIGURE 2 Model simulations compared to experimental data from isolated rat heart mitochondria. An extramitochondrial ATPase, apyrase, is titrated into the system to stimulate oxidative phosphorylation. Data and model simulations of key bioenergetics variables (NADH in A, membrane potential in B, cytochrome c^{2+} in C, extra-mitochondrial ADP concentration in D, mitochondrial pH in E, and UQH₂ in F) are shown for low [P_i] (1 mM, blue lines and symbols) and high [P_i] (5 mM, red lines and symbols) conditions. The mitochondrial pH and UQH₂ simulations are model predictions. (F, inset) Steady-state mitochondrial pH values as a function of extramitochondrial [P_i] when the ATPase rate is zero. To see this figure in color, go online.

NADH level, membrane potential ($\Delta\Psi$), fraction of cytochrome c in reduced state (cyt c^{2+}), buffer [ADP], and complex IV flux (V_{O_2}). In Fig. 2, A–D, each of the other four variables is plotted versus V_{O_2} at two different P_i concentrations. As the ATPase activity is increased, V_{O_2} increases, NADH becomes progressively oxidized, and the membrane potential is decreased. Despite a drop in NADH, the cytochrome c reduction level increases with the respiration. The model-predicted relationships between cyt c^{2+} and V_{O_2} and between $\Delta\Psi$ and V_{O_2} are different for the different P_i concentrations used in the experiments, but these relatively small differences cannot be resolved in the data based on noise in the measurements. However, there is a clear difference in the relationships between [ADP] and V_{O_2} at different P_i concentrations. Higher ATPase and V_{O_2} rates are attained at a given [ADP] with [P_i] = 5 mM compared to with [P_i] = 1 mM. This is because over the observed concentration ranges, both P_i and ADP stimulate oxidative phosphorylation in vitro.

Table 1 lists the adjustable parameters for the model used to fit the data shown in Fig. 2. The top ranked parameters in decreasing order of sensitivity are involved with substrate oxidation, ADP/ATP translocation, and extramitochondrial ATP hydrolysis. Specifically, the substrate oxidation parameters associated with the NADH/NAD⁺ mass action ratio, the maximum rate of the matrix NADH dehydrogenases, and the regulatory constants associated with NADH production, are ranked first, second, and third, respectively. The total enzyme content for ANT is the fourth highest ranked sensitive parameter. And the ADP inhibition constant for the external ATPase is the fifth highest ranking parameter. The least sensitive parameter is the activity of the endogenous proton leak. This parameter is only sensitive when the membrane potential is high and ADP is not being actively phosphorylated. Detailed definitions of these parameters in the context of the model equations are given in the Supporting Material.

Model simulations of mitochondrial pH and Q pool redox state are shown in Fig. 2, E and F. Predictions of mitochondrial pH are consistent with experimental observations (48). In addition, the pH is lower when [P_i] is high (see Fig. 2 F, inset), which is also consistent with experiments (49–51). The simulated pH values are below those reported by some studies (50) but above or in agreement with those reported by Bose et al. (11) and Baysal et al. (52). Matrix pH is highly dependent on the experimental conditions. For isolated mitochondria, when sucrose is used as the major osmotic supporting component, matrix pH is more alkaline. However, when more physiological buffers (e.g., KCl-based in the presence of permeable counterions) or body temperatures are used, the matrix is more acidic. Fig. 2 F shows that the Q pool gets progressively reduced as the workload is increased. This is because turnover of complex I is more sensitive to changes in $\Delta\Psi$ than turnover of complex III. This is partially attributed to the fact that complex I moves

TABLE 1 Adjustable Model Parameters

Parameter	Description	Value	Units	Sensitivity	Rank
X_{DH}	substrate oxidation activity (NADH-producing)	1.12×10^{-01}	mol/s/ l_{mito}	0.30	2
E_{totC1}	complex I content	1.67×10^{-04}	mol/ l_{mito}	0.05	10
E_{totC3}	complex III content	1.58×10^{-03}	mol/ l_{mito}	0.09	7
X_{C4}	complex IV activity	8.91×10^{-02}	mol/s/ l_{mito}	0.14	6
E_{totANT}	adenine nucleotide translocase content	1.41×10^{-01}	mol/ l_{mito}	0.18	4
X_{leak}	proton leak activity	1.78×10^{-03}	mol/s/ l_{mito}	0.03	11
K_{DH}	ATPase feedback constant	2.87×10^{-02}	M	0.19	3
n_{DH}	ATPase Hill coefficient	1.45	—	0.07	8
r_{DH}	effective thermodynamic NADH/NAD ⁺ ratio	35.4	—	0.30	1
α_{C2}	succinate dehydrogenase activity ($\propto$ MitoDH)	0.25	mol/s/ l_{mito}	0.06	9
K_{iADP}	K_i for ADP of external ATPase flux	2.41×10^{-04}	M	0.16	5

Local sensitivity coefficients are normalized and averaged using the following: $1/N_i \sum_{i: (df_i/dp) \neq 0} (df_i/dp)(p/f_i)$, where f_i is the i th model output, p is the parameter of interest, and N_i is the number of model outputs with nonzero df_i/dp . Model outputs are aligned with experimental data.

four protons from the matrix to the innermembrane space, while complex III only effectively pumps two protons. So when mitochondria are challenged with increasing workloads that result in depolarization of $\Delta\Psi$, the Q pool becomes more reduced because the flux through complex I is stimulated to a greater degree per mV depolarization. In addition, higher respiration rates also lead to shifts in TCA cycle intermediate concentrations and likely result in a higher fraction of electrons entering the respiratory chain from FADH₂ (53). Thus it is conceivable that SDH flux also rises with work as with NADH oxidation.

Fig. 3 explores the behavior of the model when respiration is stimulated via uncouplers rather than ATP hydrolysis. Uncoupling by FCCP is simulated by increasing proton leak activity, plotted on the x axis in Fig. 3 A. As uncoupling activity is increases, the redox poise of the cytochrome redox system becomes slightly reduced while the membrane potential decreases up to a point. The redox poise of the Q pool displays a biphasic behavior where small increases of FCCP activity lead to a reduction up until $\sim 3 \mu\text{mol H}^+/\text{s}/l_{mito}/\text{mV}$. Larger

increases beyond this value lead to a net oxidation of both the Q and cytochrome c pools. This progressive oxidation of the Q pool with increasing FCCP activity over this range is in agreement with observations by others (54,55). The Q pool redox state responds identically to respiration stimulated by uncoupling or extramitochondrial ATP hydrolysis up until the maximum capacity of OxPhos is attained (saturating extramitochondrial [ADP]). This is due to an intrinsic property of mitochondrial respiration set by the near equilibrium conditions of the ATP synthase reaction and the thermodynamic equivalency of protons whether they cross the inner mitochondrial membrane through the action of an uncoupler or the naturally occurring OxPhos-dependent proton pathways.

Calculation of the flux control coefficients for the model reveals that there is no single rate-limiting step, and there is a shift in the control coefficient profile during changes in work at low and high P_i . The flux control coefficients for the major reactions in the OxPhos pathway during increasing workloads for low and high P_i were calculated

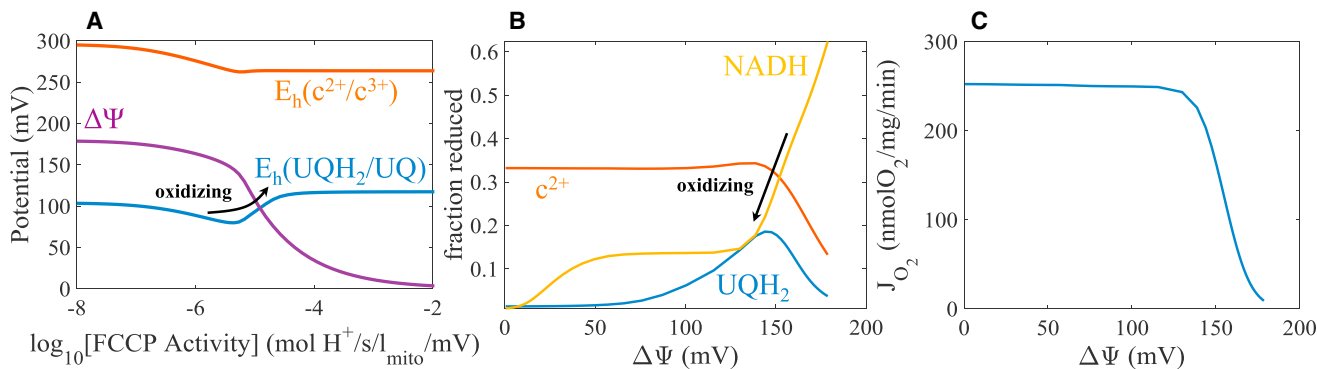


FIGURE 3 FCCP titration of mitochondrial energetics. The model is simulated with increasing amounts of FCCP to depolarize the mitochondrial membrane and stimulate respiration. (A) The redox poise of cytochrome c , ubiquinone, and $\Delta\Psi$ are shown versus FCCP activity. The effect of FCCP is modeled assuming the conductance of protons is linearly related to the FCCP concentration and the proton motive-force. (B) The fractions of reduced cytochrome c , ubiquinone, and NADH versus $\Delta\Psi$ are shown for the same conditions in (A). (C) The respiration rate is shown versus $\Delta\Psi$ for the same conditions in (A). Redox poises are computed using the following equation: $E_h = E_m - RT/zF \ln([X]_{red}/[X]_{ox})$, where E_m is the midpoint potential for a given redox couple, $[X]_{red}$ is the concentration of the reduced moiety, and $[X]_{ox}$ is the concentration of the oxidized moiety. The E_m values are set to 245 mV for the cytochrome c couple and 60 mV for the ubiquinone couple. To see this figure in color, go online.

using Eq. 1 and shown in Fig. 4. For high P_i (Fig. 4 A), the control profile is qualitatively, and in many cases quantitatively, similar to that reported previously using an experimental approach (56–59). When ATP demand is near zero and the respiration rate is low, the proton leak is the dominant respiratory component that determines O_2 consumption. But as ATP demand rises and the respiration rate increases, the control coefficient for ATPases is highest over the *in vivo* range from ~25 to 80% of maximal respiratory flux. Other contributors show increasing control coefficients as ATPase and respiratory rate are increased. Of all the respiratory complexes, complex III has the highest control coefficient when the workload is high. When $[P_i]$ is low, the control coefficients values are similar to those of the high $[P_i]$ case. However, the ATPase has a higher coefficient over a larger range for the high $[P_i]$ case compared to the low $[P_i]$ case. This is because P_i helps drive substrate oxidation (matrix dehydrogenases) via the dicarboxylate carrier and is one of the regulatory feedback components of OxPhos that sets the rate of ATP generation.

Care should be taken in interpreting the computed control coefficients, which represent the relative amount the steady-state respiratory flux would change given a change in the activity of a given enzyme or transporter. Thus, the magnitudes of the control coefficients do not necessarily reflect how the system is controlled *in vivo* or *in vitro*. It only identifies how the rate of oxygen consumption would change if a given activity were increased by a small amount. For example, for the control coefficient of the lumped dehydrogenase flux to reflect a mechanism of controlling oxidative phosphorylation, then there would have to exist a mechanism for dehydrogenase activity to increase with increasing ATP demand. More specifically, because the control coefficient for ATPase is >10 times higher than that for dehydrogenase, the activity of dehydrogenases would have to vary dramatically with ATP hydrolysis rate for dehydrogenase activity to contribute significantly to the control of respiration *in vivo*. To equal the control of feedback by ATP hydrolysis, the dehydrogenase activity would have to increase by >20-fold for every twofold increase in ATP hydrolysis.

The model simulations of steady-state hydrogen peroxide and superoxide generation for the simulations given in Fig. 2 are shown in Fig. 5. P_i has only an indirect effect through changes in $\Delta\Psi$ and ΔpH on the rate of production of free radicals. The contributions of ROS production by complexes I and III are shown in Fig. 5 A. The rate of free radical production from complex III is higher relative to complex I throughout the range of physiological workloads (approximately equal to 35–115 nmol O_2 /min/U CS). As the workload is increased, complex I derived total ROS falls with $\Delta\Psi$ and NADH, but the rate of superoxide production from complex III slightly rises due to the increase in UQH₂. Moreover, the hydrogen peroxide production rate by complex I is ~10% of the superoxide production rate, which strongly agrees with previous estimates (36). The rates of total free radical production ($2e^-$ equivalent) from complexes I and III during changes in work are shown in Fig. 5 B. In the steady state, the scavenging rate equals the production rate, so the rate of hydrogen peroxide consumption by the scavenging system equals the rate of the $2e^-$ equivalent rate of free radical production by complexes I and III. The corresponding production rates as functions of $\Delta\Psi$ are shown in Fig. 5 C. As the $\Delta\Psi$ is increased, the steady-state rate of free radical production increases and agrees with experimental observations (60).

Reports of mitochondrial ROS production as a fraction of O_2 consumption are in the range of 0.01–2% (61–63). These reported values are ROS emission rates, where ROS emission is defined as ROS production minus ROS scavenging. In the physiological setting, the fractions are more likely to be on the lower end of this range (26). Model simulations reinforce this approximate range and give values ranging from 0.1 to 2% as shown in Fig. 5 D. Note that for the simulations shown, the O_2 concentration is 200 μM , corresponding to partial pressure in room air, which is above the expected physiological range and these percentages are for total ROS generation, which includes ROS scavenging. Because ROS emission rates are underestimates of the true production rate (34,62,64,65), the simulated ROS production rates are in the range of expected values. Moreover, setting the O_2 concentration to 20 μM , a more

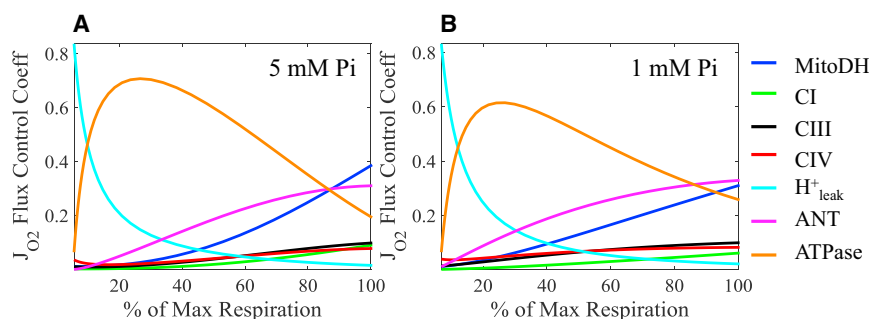


FIGURE 4 The OxPhos flux control coefficients for increasing workloads. The flux control coefficients of O_2 consumption by mitochondria under the same conditions as those for Fig. 2 are shown for high P_i (A) and low P_i (B) versus the percent of the maximum respiratory rate. The control coefficients for the independent fluxes: MitoDH, CI, CIII, CIV, proton leak, ANT, and the external ATPase are shown according to the color legend. The dependent fluxes were CII, inorganic phosphate carrier (P_iC), and F_1F_0 ATP synthase and are not calculated. For reaction close to equilibrium such as P_iC and F_1F_0 ATP synthase, the flux control coefficient is close to zero. To see this figure in color, go online.

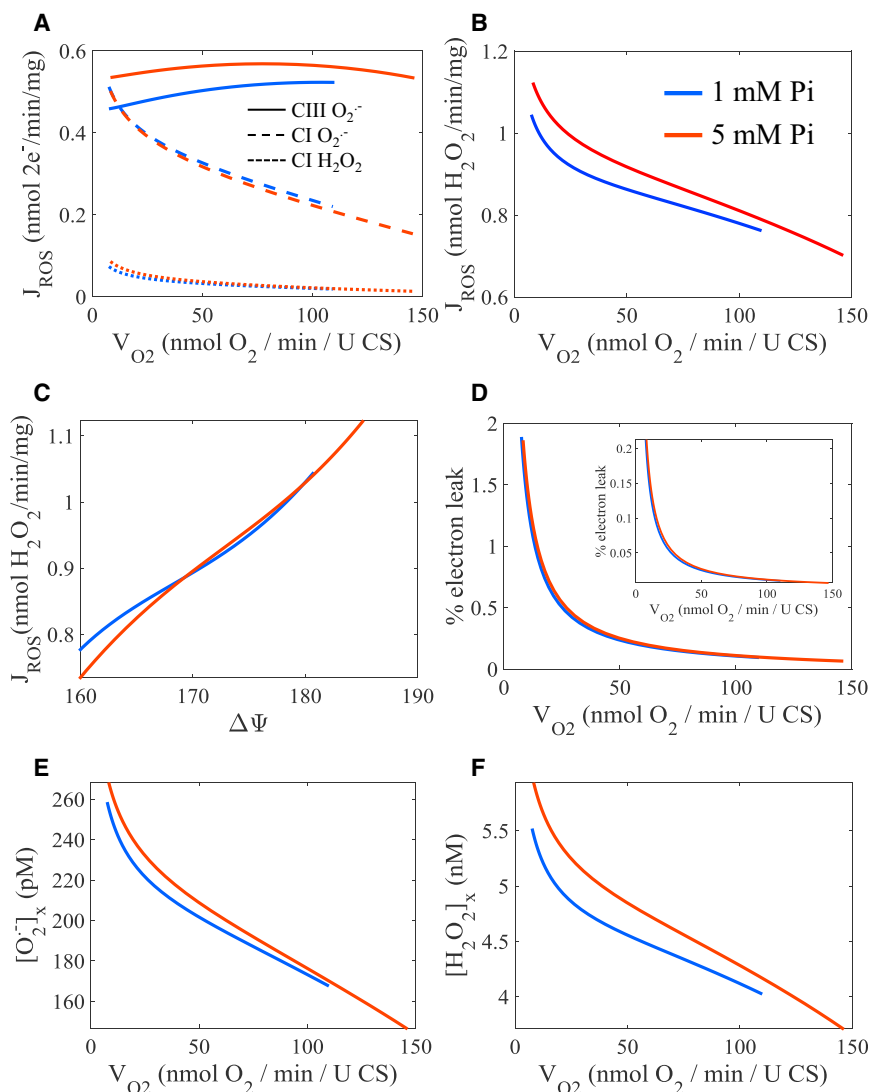


FIGURE 5 Model simulations of ROS steady-state ROS production, scavenging, and concentrations. (Blue lines) Low $[P_i]$ (1 mM) case; (red lines) high $[P_i]$ (5 mM) case. The conditions are as given in Fig. 2. (A) The individual contributions of ROS production by complexes I and III are shown for the apyrase titration. (B) The steady-state ROS production rates are shown versus the apyrase titration. (C) The rate of ROS production versus $\Delta\Psi$ is shown. (D) The % of electrons that leak to form ROS instead of H_2O is shown versus the apyrase titration. (Inset) Simulation results with a more physiological O_2 concentration of $20\ \mu M$. (E and F) The steady-state concentrations of superoxide and H_2O_2 inside the mitochondria are shown versus the apyrase titration. To see this figure in color, go online.

physiological value, results in an order-of-magnitude drop in free radical production rates relative to O_2 consumption (<0.01 – 0.2% , inset of Fig. 5 D). Therefore, one must take care when attempting to extrapolate in vitro experiments to in vivo conditions. Fig. 5, E and F, shows the steady-state concentrations of superoxide and H_2O_2 during changes in workload. These values are in agreement with those reported (66–68).

Simulation results in Fig. 5 are under conditions of forward electron transport, where electrons are transferred along the ETC from NADH and $FADH_2$ to O_2 . Mitochondrial can produce free radicals at 10-fold higher production rates during reverse electron transport (RET) conditions, which is a pathological mode of superoxide production that requires both a high $\Delta\Psi$ and highly reduced Q pool (see below). During RET, mitochondrial ROS production is much higher relative to O_2 consumption and is responsible for the burst of ROS seen during reperfusion (43).

Using the model, we explored the so-called bistability phenomenon reported to occur upon reperfusion after an ischemia or hypoxic period (69–71). This phenomenon is characterized by two distinct rates of free radical production from complexes I and III for the same or very similar conditions, which was proposed to explain the different ROS production rates before and after a perturbation such as ischemia/reperfusion. The model was appended to include the ATPase flux expression and phosphoenergetics dynamics of Wu et al. (1) that include the creatine kinase and adenylate kinase reactions and the extramitochondrial water space to mitochondrial volume ratio was adjusted in accordance with measurements of cardiomyocyte density and composition data (42). The details are given in the Supporting Material. Fig. 6 shows the model results for a simulated ischemia/reperfusion protocol. For the initialization period, the model parameters given in Table 1 and the Supporting Material were used. For the ischemic period, the O_2

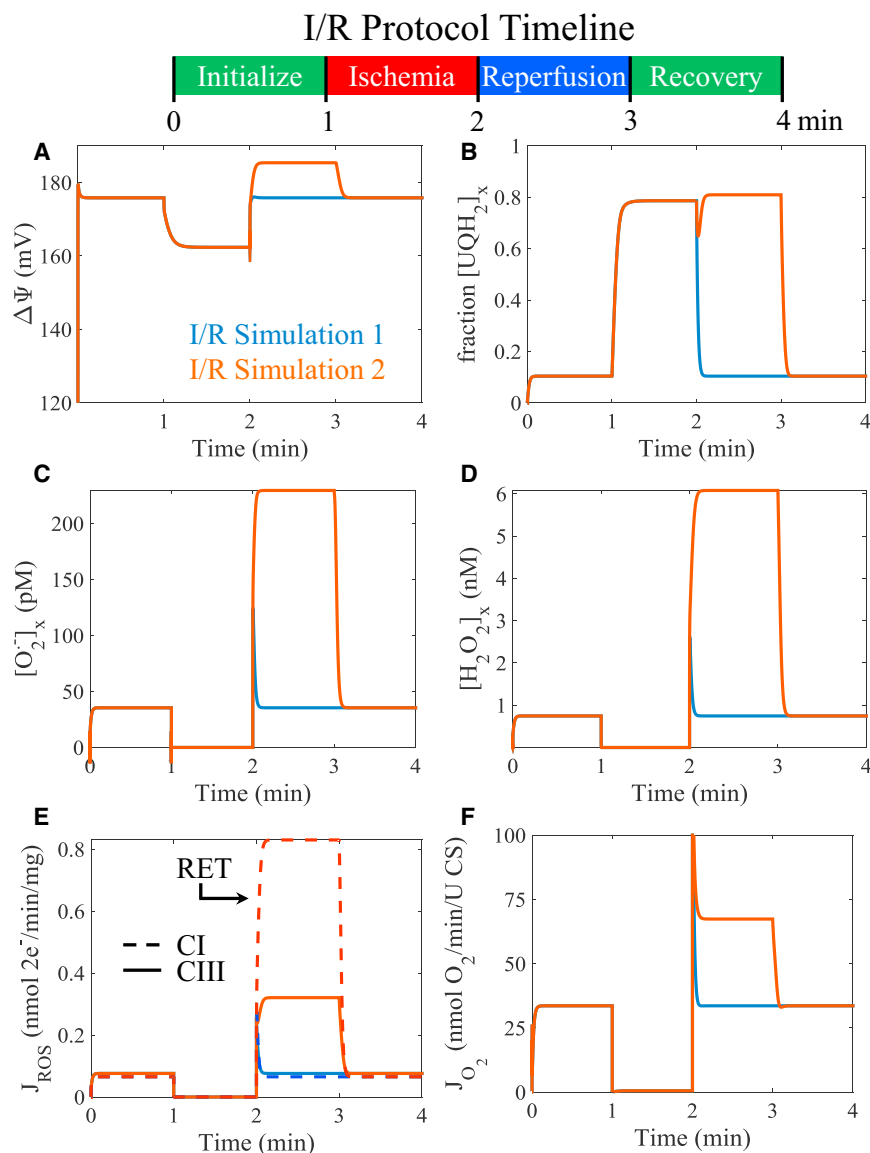


FIGURE 6 Model simulations of ischemia/reperfusion (I/R). Two I/R simulations are run for the I/R protocol timeline shown. The model parameters are given in [Table 1](#) and the [Supporting Material](#). For the initialization period, the O_2 concentration is set to $30\ \mu M$, and the ATPase activity is set to $2\ mM/s$ to produce a moderate workload demand. For the ischemic period, the O_2 concentration is set to $1\ nM$ to reflect the hypoxic conditions during ischemia. For the reperfusion period, the O_2 concentration is reset to its original value for both I/R Simulations 1 and 2. For I/R Simulation 2, the complex II activity and its apparent equilibrium constant are both increased by a factor of 10 to account for an accumulation of succinate that occurs during ischemia (72,73). In the recovery period, the model parameters are reset to the conditions for the initialization period. The steady-state model output for $\Delta\Psi$, fraction of UQH_2 , superoxide, H_2O_2 , ROS production and O_2 consumption are shown for (A)–(F), respectively. To see this figure in color, go online.

concentration was set to $1\ nM$ to reflect the hypoxic conditions during ischemia. For the reperfusion period, the O_2 concentration was set back to the baseline value.

Two different ischemia/reperfusion (I/R) simulations were performed. For I/R Simulation 1, the normal mitochondrial model was used for the full time course. For I/R Simulation 2, the complex II activity and its apparent equilibrium constant were both increased upon reperfusion to account for an accumulation of succinate that occurs during ischemia (72,73). This modification is further justified by an analysis of the model behavior during ischemia. As shown in [Fig. 6 A](#), $\Delta\Psi$ does not fully depolarize during ischemia despite the absence of O_2 . Flux through complexes III and IV goes to zero, but complex I flux does not (not shown). This is due to reversal of complex II. Complex II supplies complex I with UQ so that it maintains the $\Delta\Psi$ at a cost of accumu-

lating succinate. In the recovery period, the model parameters are reset to the conditions for the initialization period. Simulating ischemia/reperfusion using I/R Simulation 1 did not lead to a bistability in ROS generation and the system returns to its initial operating point during the reperfusion period. But I/R Simulation 2 does produce a bistability in ROS generation because the rate of UQH_2 production was greater after the ischemic/hypoxic period. Furthermore, the $\Delta\Psi$ during the reperfusion period is slightly hyperpolarized for I/R Simulation 2 because the increase in complex II flux leads to higher flux through complexes III and IV. RET was maintained by a highly reduced Q pool as shown in [Fig. 6 B](#), which led to elevated superoxide and H_2O_2 , as shown in [Fig. 6 B and C](#). The individual contributions to ROS production by complexes I and III are shown in [Fig. 6 E](#). For the I/R Simulation 2 conditions, RET is the dominant source of ROS

and leads to a 10-fold higher ROS production rate compared to forward electron transport (as compared to ROS production during the initialization period). This is only possible when there is a buildup of substrates that directly lead to UQH₂ (e.g., succinate). The O₂ consumption rates are also higher for I/R Simulation 2 as shown in Fig. 6 F due to a combination of higher UQH₂ and cyt c²⁺ (not shown).

Fig. 6 presents a simplified simulation of a very complex phenomenon. There are several additional enzymes capable of producing ROS that are not included in this model formulation (33,62). These sources are important under certain conditions and may be added to expand the model to include more detailed free radical scavenging dynamics (74–76). Heterogeneities, transport, and diffusion-limited processes are also not modeled, so model dynamics do not reflect exactly what is anticipated to occur in vivo. For example, the dynamics of free radical production in ischemic tissues is on the order of minutes (77–79), but these include a variety of extramitochondrial factors that are beyond the scope of this study. In the model simulations, the O₂ concentration was instantaneously changed to induce changes in state-variable dynamics. In a more realistic simulation, the rapid changes shown in Fig. 6 would be slowed. However, it does demonstrate that to generate a burst of ROS during reperfusion after a period of ischemia or hypoxia, an additional fuel source that directly feeds electrons into the Q pool must be available. This conclusion is strongly corroborated by Chouchani et al. (43), who demonstrate that elevated ROS production during reperfusion after ischemia is due to accumulation of mitochondrial succinate. During ischemia, there is partial reversal of the malate/aspartate shuttle and fumarate overload caused by purine nucleotide breakdown, which leads to elevated mitochondrial succinate levels that lead to an over-reduced Q pool. Upon reperfusion, the reintroduction of O₂ reenergized the mitochondrial membrane and restarted oxidative ATP synthesis. It is at this point when the excess succinate produced during ischemia delivers a deleterious amount of ROS via RET from complex I as well as from complex III. The model simulations support this sequence of events.

In summary, we have presented here an oxidative-phosphorylation model that is capable of simulating ROS dynamics is presented. The model includes updated descriptors for complexes I, III, and IV, and ANT. It is fit to recent data obtained from isolated rat heart mitochondria subjected to physiological conditions and changes in workload. The model reproduces the data with high fidelity and predicts that an increase the Q-pool redox state is responsible for the apparent P_i activation of complex III. In addition, the steady-state simulation of hydrogen peroxide and superoxide production and levels are in agreement with the experimental data. It is shown that complex III is responsible for the more ROS production during changes in workload relative to complex I. However, this relationship is reversed under pathological conditions. Finally, the model is used to

investigate how elevated levels of succinate are responsible for the burst of free radicals seen during reperfusion after an ischemic/hypoxic period.

SUPPORTING MATERIAL

Supporting Materials and Methods, three figures, and seventeen tables are available at [http://www.biophysj.org/biophysj/supplemental/S0006-3495\(16\)00035-7](http://www.biophysj.org/biophysj/supplemental/S0006-3495(16)00035-7).

AUTHOR CONTRIBUTIONS

J.N.B. designed research, performed research, contributed analytic tools, analyzed data, and wrote the article; D.A.B. designed research and wrote the article; and K.C.V. designed research, performed research, analyzed data, and wrote the article.

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